HYBRID ADAPTIVE TRANSFORMER DIFFERENTIAL PROTECTION:

AN INTELLIGENT DWT-LSTM FRAMEWORK

STUDENT NAME

B.Sc. ENGINEERING THESIS

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY

DHAKA, BANGLADESH

JANUARY 2026

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STUDENT NAME (SN. XXXXXXXXXX)

A Thesis Submitted in Partial Fulfillment of the Requirements for the

Degree of Bachelor of Science in Electrical and Electronic Engineering

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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APPROVAL CERTIFICATE

This is to certify that the thesis entitled

"Hybrid Adaptive Transformer Differential Protection:

An Intelligent DWT-LSTM Framework"

submitted by Student Name (SN. XXXXXXXXXX) to the Department of Electrical and Electronic Engineering, Military Institute of Science and Technology (MIST), Mirpur Cantonment, Dhaka, Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science in Electrical and Electronic Engineering, has been accepted as satisfactory for the award of the degree and approved by the undersigned examiners.

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Counter-signed on behalf of the Board of Advanced Studies

DECLARATION

I hereby declare that the thesis entitled

“Hybrid Adaptive Transformer Differential Protection:

An Intelligent DWT-LSTM Framework”

submitted to the Department of Electrical and Electronic Engineering, Military Institute of Science and Technology (MIST), Mirpur Cantonment, Dhaka, Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science in Electrical and Electronic Engineering, is a record of original work carried out by me under the supervision of Supervisor Name, PhD, Professor, Department of EEE, University. I hereby affirm that:

This thesis has not been submitted elsewhere, in whole or in part, for the award of any degree, diploma, or fellowship at this or any other university, institute, or organization.

The work presented herein is entirely the original research and intellectual contribution of the candidate, except where explicit reference has been made to the work of other researchers.

Proper acknowledgment has been made in the text to all other materials used, and all sources of information have been duly cited following the IEEE referencing format.

No part of this thesis has been fabricated, plagiarized, or misrepresented in any manner, and the results presented are accurate to the best of my knowledge and belief.

I have complied with all ethical guidelines and academic integrity standards set forth by the Military Institute of Science and Technology.

Candidate's Signature: ________________________________

Name: Student Name

Student Number: XXXXXXXXXX

Date:

DEDICATION

Dedicated to my parents and family

for their unwavering love and support

throughout my academic journey.

ACKNOWLEDGEMENTS

First and foremost, all praises are due to Almighty Allah for His infinite blessings, guidance, and mercy, without which this endeavor would not have been possible. I am profoundly grateful for the strength, patience, and wisdom He bestowed upon me throughout the course of this research and my entire academic life.

I would like to express my deepest and most sincere gratitude to my thesis supervisor, Supervisor Name, PhD, Professor, Department of Electrical and Electronic Engineering, for his invaluable guidance, continuous encouragement, and unwavering support throughout every phase of this research. His profound expertise in power system protection, insightful suggestions, and patient mentorship have been instrumental in shaping the direction and quality of this work.

My sincere thanks go to Head of Department Name, Head of the Department of Electrical and Electronic Engineering, MIST, for providing the necessary academic facilities, laboratory resources, and an excellent research environment. I am also grateful to all faculty members of the Department of EEE at MIST whose teachings in power systems, signal processing, and machine learning directly contributed to the interdisciplinary nature of this research.

I extend my heartfelt appreciation to my fellow classmates, laboratory mates, and colleagues at MIST. Their camaraderie, intellectual discussions, technical assistance with MATLAB/Simulink simulations, and willingness to share knowledge in deep learning and signal processing have been invaluable.

I am profoundly indebted to my parents, whose sacrifices, unconditional love, and steadfast encouragement have been the bedrock of my educational pursuits. I also extend my deepest gratitude to my siblings, extended family members, and close friends who provided emotional support throughout this demanding academic journey.

Finally, I acknowledge with gratitude the infrastructure and computational resources provided by MIST, as well as the open-source contributors and researchers whose published works in power system protection, wavelet analysis, and deep learning have served as the foundational pillars upon which this research is built.

Date:

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ABSTRACT

Power transformers constitute the most critical and capital-intensive components of electrical power transmission and distribution networks, making their reliable and rapid protection a paramount concern for ensuring the continuity and stability of electric power supply . Differential protection has long been recognized as the primary and most effective scheme for safeguarding power transformers against internal faults, operating on the fundamental principle of comparing the currents entering and leaving the protected zone . However, the conventional percentage differential relay faces significant challenges in accurately distinguishing between genuine internal fault conditions and non-fault transient events such as magnetizing inrush currents, sympathetic inrush, and external faults with current transformer (CT) saturation . These phenomena produce substantial differential currents that closely resemble those generated by actual internal faults, frequently leading to false tripping. The second harmonic restraint method has been shown to suffer from reduced sensitivity in modern transformers employing high-permeability core materials that produce lower levels of second harmonic content during energization .

The emergence of digital signal processing techniques and computational intelligence has opened promising avenues for developing more sophisticated and reliable transformer protection algorithms . In particular, the Discrete Wavelet Transform (DWT) has proven to be an exceptionally powerful tool for the analysis of non-stationary transient signals, offering superior time-frequency localization capabilities compared to the traditional Fourier transform . Simultaneously, Long Short-Term Memory (LSTM) recurrent neural networks have demonstrated remarkable performance in learning complex temporal dependencies and sequential patterns from time-series data, making them exceptionally well-suited for power system protection tasks that inherently involve temporal signal analysis .

This thesis proposes a novel hybrid adaptive protection framework that synergistically integrates the multi-resolution feature extraction capabilities of the DWT with the advanced temporal classification power of LSTM networks. The proposed methodology employs a three-level wavelet decomposition using the Daubechies-4 (db4) mother wavelet, selected for its jagged, asymmetrical shape that optimizes detection of fault inception transients . From the decomposed wavelet coefficients, six energy features are extracted per time step based on Parseval's theorem: the approximation energy and detail energy for each of the three phases (A, B, and C). These features are computed over a half-cycle sliding window of 16 samples at a sampling frequency of 1.6 kHz (32 samples per cycle at 50 Hz) . The trained LSTM model is exported to the ONNX format (Opset 14) and integrated into the MATLAB/Simulink environment via the Deep Learning Predict block for real-time protection validation.

The classification module employs a multi-layer LSTM network architecture with a global temporal attention mechanism, designed to discriminate between four distinct operating states: normal steady-state operation, external fault conditions, magnetizing inrush conditions, and internal fault conditions . The LSTM network leverages its internal gating mechanisms to selectively retain and propagate relevant temporal information across the 32-sample sliding window . A critical innovation of this thesis is the hybrid decision handshake architecture, wherein the LSTM classifier acts as a supervisory veto (AND) gate for the classical adaptive 87T differential relay logic. The relay only trips if both the adaptive 87T differential element operates and the LSTM classifies the event as an internal fault with a confidence exceeding a predefined threshold.

A comprehensive simulation model of a 300 MVA, 230 kV/11 kV power transformer with a Yd1 vector group configuration is developed using MATLAB/Simulink with the Simscape Electrical toolbox for generating realistic training and testing datasets . The discrete-time simulation employs a solver sample time of 1×10^{-5} s (10 μs) to capture high-frequency transients and nonlinear magnetic behavior. Extensive simulation scenarios encompass a wide range of fault types, fault inception angles, fault locations, fault resistances, loading conditions, CT saturation conditions, remanent flux levels, and external fault conditions. Automated batch data generation is performed through a MATLAB control script (ControlPanel.m). The generated differential current waveforms are processed through the DWT engine, and the extracted feature vectors are used to train the LSTM classifier in Python using PyTorch, followed by ONNX export for Simulink integration.

The proposed hybrid DWT-LSTM framework demonstrates exceptional performance across all evaluation metrics. The classification model achieves zero false negatives (100% dependability) across all internal fault scenarios, with an overall relay operating time well within the sub-cycle requirement (less than 20 ms at 50 Hz) . The ONNX inference latency in Simulink is measured at less than 0.74 ms per prediction step, demonstrating real-time feasibility. The hybrid veto logic eliminates all false trips (the 77 false trips otherwise issued by the standalone adaptive 87T relay during external faults with deep CT saturation and sympathetic inrush conditions on the test set), achieving 100% security. The framework exhibits robust generalization under extreme stress conditions, including CT remanent flux mismatches up to ±95%, frequency deviations from 47 Hz to

53 Hz, and noise injection down to 20 dB SNR .

The principal contributions of this thesis include: (i) a systematic methodology for power transformer protection simulation using MATLAB/Simulink, encompassing detailed physical modeling with nonlinear CT saturation and transformer core saturation; (ii) the design of a compact yet highly effective six-dimensional energy feature vector derived from three-level db4 wavelet decomposition; (iii) a hybrid decision architecture where the LSTM serves as a supervisory veto for the classical adaptive 87T relay, achieving zero false negatives while eliminating false trips; (iv) the complete real-time deployment pipeline from Python/PyTorch training through ONNX export to the Simulink Predict block; and (v) extensive stress testing and out-of-distribution validation. The results demonstrate that the proposed hybrid DWT-LSTM framework offers a highly reliable, fast-acting, and practically deployable solution for next-generation intelligent transformer differential protection systems.

Keywords: Transformer differential protection, Discrete Wavelet Transform (DWT), Long Short-Term Memory (LSTM), MATLAB/Simulink, magnetizing inrush current, CT saturation, ONNX, real-time relay, fault classification, deep learning, hybrid adaptive protection, intelligent relaying.

CONTENTS

(Right-click and choose “Update Field” in Word to regenerate page numbers.)

LIST OF ABBREVIATIONS

LIST OF TABLES AND FIGURES

Note: Numbered table and figure captions appear in-text throughout the chapters. Dashed placeholder boxes mark every figure and data table to be inserted; replace each box with the corresponding image or completed table, then insert a List of Tables and List of Figures field in Word and update cross-references.

CHAPTER 1 INTRODUCTION

1.1 Background

Power transformers are among the most critical and capital-intensive assets in electrical power systems, serving as the essential link between generation, transmission, and distribution networks . An unscheduled outage incurs enormous direct replacement costs and indirect losses from energy not served and potential cascading failures . Differential protection, grounded in Kirchhoff’s Current Law (KCL), remains the primary protection scheme. Under healthy conditions, the input (primary) current and output (secondary) current are related by the turns ratio n:

When an internal fault occurs, this balance is disrupted and the differential current becomes significantly nonzero, triggering relay operation .

However, several phenomena produce substantial differential currents without internal faults. Magnetizing inrush current, arising during transformer energization, can reach 5–15 times the rated current with significant even-harmonic content . Current transformer (CT) saturation distorts secondary waveforms under large DC-offset fault currents , and overexcitation conditions introduce additional harmonic components . Traditional harmonic restraint (HR) and harmonic blocking (HB) methods face increasing limitations as renewable energy sources, power electronic converters, and advanced core materials alter the waveform signatures that conventional relays depend upon . This thesis proposes a hybrid adaptive framework integrating the discrete wavelet transform (DWT) with long short-term memory (LSTM) networks for more robust discrimination between fault and non-fault conditions .

1.2 Research Motivation

The reliability of differential protection is increasingly threatened by multiple compounding factors. First, conventional harmonic restraint relies on fixed second-harmonic thresholds (typically 15–20%), yet modern transformers with amorphous alloy cores exhibit reduced second harmonic content during inrush that can fall below these thresholds , while internal faults near voltage zero-crossings can generate sufficient harmonics to inadvertently restrain the relay . Second, CT saturation produces asymmetric distortion that creates spurious differential currents persisting for several cycles , and legacy CTs with degraded performance further elevate this risk .

Third, advanced core materials such as amorphous metals and high-B-saturation alloys exhibit steeper B-H curves with altered harmonic signatures . Fourth, renewable energy converters inject non-sinusoidal currents with limited fault contributions (1.1–1.5 p.u.), fundamentally changing differential current signatures . Fifth, complex configurations—multi-winding transformers, autotransformers, and on-load tap changers—narrow the margin between legitimate restraint and genuine fault conditions . These challenges motivate a new generation of protection algorithms leveraging advanced signal processing and computational intelligence .

1.3 Problem Statement

Despite over a century of development, transformer differential protection schemes continue to experience maloperations at unacceptable rates. The problem addressed in this thesis can be formally stated as follows.

Given: time-series differential current signals $\{I_d(t)\}$ sampled at f_s samples per second, encompassing internal faults, magnetizing inrush, external faults with CT saturation, overexcitation, and normal load conditions; signals corrupted by measurement noise, CT transformation errors, and potential communication delays.

Find: a classification function $f : X \rightarrow Y$ mapping the input signal x X to output classes $Y = \{y_{fault}, y_{inrush}, y_{external}, y_{normal}\}$ with high accuracy and reliability.

Subject to: operating time within a half-cycle window (10 ms at 50 Hz); dependability with missed-detection probability $P_{miss} \leq 0.01$; security with false-trip probability $P_{false} \leq 0.01$; robustness across fault levels, loading conditions, and operating states; and generalizability to unseen operating conditions not represented in the training data.

The key difficulties arise from the highly nonlinear and non-stationary feature space, context-dependent classification boundaries, real-time computational constraints, and scarcity of real-world fault data . Conventional harmonic restraint methods rely on fixed thresholds ; waveform-based methods are noise-sensitive; traditional machine learning approaches (ANN, SVM) lack temporal dependency modeling ; and basic CNNs or RNNs either fail to capture sequential patterns or suffer from vanishing gradients .

1.4 Research Objectives

The overarching goal is to design and validate a hybrid adaptive differential protection scheme based on a DWT-LSTM framework using MATLAB/Simulink, pursued through four objectives:

Simulation Model Development and Data Generation. Develop a detailed electromagnetic-transient (EMT) simulation model of a power transformer protection system in MATLAB/Simulink, generating differential current signals for internal faults (SLG, LL, LLL, inter-turn), magnetizing inrush (including sympathetic inrush), external faults with CT saturation, overexcitation, and normal load operation .

Optimal DWT Feature Extraction. Optimize the application of DWT for multi-resolution decomposition, identifying the optimal mother wavelet, decomposition level, and feature set that maximize fault/non-fault separability, through systematic comparison of wavelet families .

Hybrid DWT-LSTM Classification and Adaptive Decision Logic. Design a hybrid architecture integrating wavelet-based feature extraction with LSTM temporal sequence modeling, optimizing network hyperparameters, and develop adaptive decision-making with dynamic thresholds based on real-time operating conditions and LSTM confidence scores .

Performance Evaluation, Benchmarking, and Robustness Analysis. Benchmark the proposed framework against conventional harmonic restraint, waveform-based, ANN, SVM, and CNN methods using standardized metrics, and assess robustness under measurement noise, CT errors, frequency deviations, and distribution shift .

1.5 Scope and Limitations

1.5.1 Scope of the Work

Transformer Model: a two-winding 300 MVA, 230/11 kV transformer with Yd1 vector group is modeled in detail; generalization to other ratings and vector groups (e.g., Dy11, YNynd0) and voltage levels is identified as future work.

Fault Types: SLG, LL, LLL, LLLG, and inter-turn faults (5%, 10%, 20% of winding); external faults for security assessment.

Signal Processing: one-dimensional DWT decomposition; CWT and WPT are discussed but not implemented.

Machine Learning: LSTM as primary classifier with ANN, SVM, and CNN baselines; transfer learning and ensemble methods are excluded.

Validation: simulation-generated data in MATLAB/Simulink; hardware-in-the-loop and RTDS validation are deferred to future work.

Frequency: nominal 50 Hz system; adaptable to 60 Hz but configured for 50 Hz.

1.5.2 Limitations

Training data is simulation-generated; discrepancies with real-world field data may exist due to unmodeled dynamics and manufacturing tolerances.

No validation on physical relay hardware; computational analysis is based on software benchmarks.

Class imbalance is mitigated via data augmentation, but generalization to real distributions requires further validation.

The adaptive threshold mechanism is designed for static loading; dynamic scenarios (renewable ramps, motor starting) are not explicitly addressed.

Communication failures in IEC 61850 digital substation architectures are not considered.

1.6 Contributions of This Work

Contribution 1 — A Systematic Wavelet Feature Selection Framework. A quantitative methodology for selecting the optimal mother wavelet, decomposition level, and feature extraction strategy using Fisher's discriminant ratio and mutual information, ensuring maximum inter-class separation .

Contribution 2 — A Novel Hybrid DWT-LSTM Architecture. A serial fusion architecture where the DWT module decomposes differential currents into time-frequency sub-band features that are sequentially fed into a stacked LSTM network, combining localized time-frequency decomposition with long-term temporal dependency modeling. A comprehensive benchmarking dataset of over 10,000 simulated events evaluated across seven methods using standardized metrics is also provided .

Contribution 3 — An Adaptive Decision Framework with Robustness Validation. An adaptive decision logic with dynamic thresholds based on LSTM confidence scores, differential/restraint current magnitudes, and system operating state, incorporating a probabilistic layer for classification uncertainty . Detailed robustness analysis under noise (20–60 dB SNR), CT ratio errors (±5%), frequency deviations, and distribution shift demonstrates graceful performance degradation well above practical deployment thresholds .

1.7 Thesis Organization

This thesis is organized into seven chapters. Chapter 1 provides the background, motivation, problem statement, objectives, scope, and contributions. Chapter 2 presents a critical literature review and identifies the research gaps. Chapter 3 establishes the mathematical foundations of differential protection, inrush phenomena, CT saturation, and classification theory. Chapter 4 details the MATLAB/Simulink EMT simulation model and the generated dataset. Chapter 5 presents the proposed DWT-LSTM methodology. Chapter 6 presents experimental results, comparative benchmarking, and robustness analysis. Chapter 7 summarizes the key findings and outlines future directions.

CHAPTER 2 LITERATURE REVIEW

2.1 Introduction to Literature Review

This chapter presents a critical review of the existing literature on transformer differential protection, spanning conventional harmonic restraint and waveform-based methods, signal processing techniques (Fourier and wavelet transforms), machine learning approaches (ANN, SVM, PNN, ensemble methods), and deep learning architectures (CNN, RNN, LSTM). The review identifies the strengths and limitations of each category and highlights the specific research gaps that motivate the hybrid DWT-LSTM framework proposed in this thesis. The surveyed literature is drawn primarily from IEEE Transactions on Power Delivery, IEEE Transactions on Power Systems, Electric Power Systems Research, and related conference proceedings published from the 1950s through 2024.

2.2 Conventional Transformer Differential Protection Methods

2.2.1 Harmonic Restraint Methods

The harmonic restraint (HR) method, first proposed by Sharp and Glassburn in 1958, is the most widely deployed technique for discriminating magnetizing inrush from internal faults . The premise is that inrush currents contain significant even-harmonic content (particularly the second harmonic, with $k2 = I2/I1$ typically ranging from 15% to 60%), while internal fault currents are predominantly fundamental and odd harmonics. The relay trips only when the following conditions are simultaneously satisfied:

$I\_d > I\_pickup$

$I\_d > S \cdot I\_r$

where I_d is the differential current, I_r the restraint current, I_pickup the minimum pickup current, S the slope setting, and k_threshold is typically set between 15% and 20%.

Despite widespread adoption, HR suffers from documented shortcomings: modern transformers with amorphous alloy cores exhibit second harmonic ratios as low as 7–10%, below the conventional 15% threshold, causing false tripping ; internal faults near voltage zero-crossing produce DC offsets with sufficient second harmonic to inadvertently restrain the relay ; and cross-blocking schemes improve security during three-phase energization but increase the risk of failure to trip for simultaneous inrush and fault events .

2.2.2 Waveform-Based and Equivalent Circuit Methods

Waveform-based methods exploit the distinct shape of inrush versus fault current waveforms. The dead-angle method defines the angular duration during each half-cycle where the current magnitude falls below a threshold; if this exceeds 65–80 degrees, the event is classified as inrush. While computationally straightforward, dead-angle methods are highly sensitive to threshold selection, measurement noise, and CT saturation . Gap-detection methods operate on the current derivative and are more noise-susceptible , while waveform-correlation methods compute half-cycle symmetry but are sensitive to data-window length and CT saturation .

Equivalent circuit-based methods model the transformer using magnetic equivalent circuits, detecting internal faults when terminal measurements violate the circuit equations . Voltage-restrained differential protection increases relay restraint during overexcitation based on fifth-harmonic content . While physically grounded, these methods require accurate transformer models, voltage measurements, and real-time solutions to differential equations, limiting their practical deployment .

2.3 Signal Processing Techniques

2.3.1 Fourier and Wavelet Transform Approaches

The Discrete Fourier Transform (DFT) has been the workhorse of digital relaying since the 1980s. Full-cycle DFT provides excellent frequency selectivity but incurs a one-cycle delay, while half-cycle DFT is faster but susceptible to decaying DC components . The fundamental limitation of Fourier methods is the uncertainty principle: the DFT provides frequency resolution at the expense of time-domain localization, making it unable to capture the transient, non-stationary characteristics of inrush and fault waveforms .

The wavelet transform overcomes this limitation through simultaneous time-frequency localization with adaptive resolution. For digital implementation, the Discrete Wavelet Transform (DWT) is preferred, using cascaded low-pass and high-pass filters with downsampling to produce multi-resolution decomposition . In , DWT with db4 at five levels achieved 97.3% accuracy with an ANN classifier; proposed Wavelet Packet Transform (WPT) with sym8 for improved entropy-based discrimination; and concluded that db4 provides an optimal trade-off for 50 Hz signals. A persistent limitation of wavelet-based methods is that they extract features from individual windows without modeling temporal evolution across successive windows.

2.4 Machine Learning Approaches

Artificial Neural Networks (ANNs). Backpropagation-trained multilayer perceptrons have been extensively applied to classify differential current events. A three-layer MLP with harmonic features achieved 98.5% accuracy on 2,000 simulated events , while incorporating wavelet energy and entropy features improved accuracy to 99.1% . However, ANNs process features as static vectors, discarding temporal ordering information critical for transient discrimination.

Support Vector Machines (SVMs). SVMs with RBF kernels achieved 98.2% accuracy using wavelet features, with superior generalization from limited data . While robust to noise via margin maximization, their cubic training-cost scaling and inherent binary formulation limit applicability to multi-class fault discrimination, and they treat inputs as static feature vectors without temporal modeling.

Probabilistic Neural Networks (PNNs). PNNs employ non-iterative Parzen-window density estimation for Bayesian classification, achieving 97.8% accuracy with harmonic features and 98.9% with combined wavelet features . However, computational and memory requirements scale linearly with training samples, and performance is sensitive to the smoothing parameter.

Ensemble and Hybrid Methods. Random Forest classifiers using combined Fourier and wavelet features achieved 99.0% accuracy with built-in feature-importance measures . Two-stage hybrid schemes reduced computational burden while maintaining accuracy , at the cost of increased training complexity and reduced interpretability.

2.5 Deep Learning Approaches

2.5.1 CNN, RNN and LSTM Architectures

Convolutional Neural Networks (CNNs) have been adapted for 1-D time-series classification of differential currents. A 1D-CNN processing raw waveforms at 1 kHz over two-cycle windows achieved 98.7% accuracy, outperforming ANNs with hand-crafted Fourier features (96.3%) . However, CNNs treat the input as a fixed-length vector and do not explicitly model global temporal dependencies across successive cycles—information critical for distinguishing events with evolving characteristics.

Recurrent Neural Networks (RNNs) address this limitation by maintaining a hidden state updated at each time step, but standard RNNs suffer from the vanishing-gradient problem. The Long Short-Term Memory (LSTM) network overcomes this through a gated memory cell with input gate i_t, forget gate f_t, and output gate o_t:

where σ is the sigmoid function and denotes the Hadamard product. An LSTM classifier processing sequences of wavelet-extracted features over successive half-cycle windows achieved 99.3% accuracy , significantly outperforming ANNs and CNNs. Attention mechanisms have been integrated with LSTM to assign learned importance weights to different time steps, achieving high accuracy with reduced complexity and interpretable attention weights . A comprehensive comparison of vanilla RNN, GRU, and LSTM concluded that LSTM consistently outperforms alternatives for complex transient scenarios .

2.5.2 Emerging Deep Learning Architectures

Beyond CNNs and RNNs, autoencoders have been explored for unsupervised anomaly detection through reconstruction-error analysis , and Generative Adversarial Networks (GANs) have been proposed for synthetic fault and inrush data augmentation . The Transformer architecture, with its self-attention mechanism, has been adapted for power-system time-series classification but remains in early stages for transformer protection, with quadratic computational scaling posing real-time implementation challenges .

2.6 Research Gaps

Despite extensive literature, five critical gaps motivate this thesis. (1) Wavelet parameter selection has been ad hoc rather than systematically optimized using quantitative separability metrics. (2) Existing approaches either use wavelet features as static inputs to feedforward classifiers (discarding temporal information) or feed raw waveforms to sequence models (discarding multi-resolution decomposition); no prior work fully integrates DWT feature extraction with LSTM temporal modeling in a serial hybrid architecture. (3) The vast majority of ML/DL methods employ fixed decision thresholds that do not adapt to real-time operating conditions. (4) Most studies lack comprehensive robustness evaluation under realistic adversities. (5) The absence of standardized datasets, metrics, and benchmarking protocols hinders fair cross-method comparison. This thesis addresses all five gaps through systematic wavelet optimization, a novel DWT-LSTM hybrid architecture, adaptive dynamic thresholding, comprehensive robustness analysis, and standardized benchmarking against seven baseline methods.

2.7 Comparative Synthesis of the Literature

The preceding sections reviewed individual methodological families. To consolidate this body of work and to position the present thesis precisely against the state of the art, Table 2.1 synthesises the most directly comparable studies drawn from the reference list. For each work, the table records the signal-processing front end, the classifier or decision mechanism, the principal feature set, the best reported accuracy (as stated by the respective authors and measured on their own datasets), the validation modality, and the dominant residual limitation that the present work seeks to address. Because each study uses a different transformer model, dataset size, and evaluation protocol, the reported accuracies are not directly commensurable; they are reproduced here only to characterise the maturity of each approach, not to rank methods against one another. A controlled, like-for-like comparison on a single unified dataset is deferred to Chapter 6 (Section 6.8), where every baseline is re-implemented and evaluated on the same 12,800-case corpus used for the proposed framework.

Three observations follow from this synthesis. First, the field has progressed from fixed-threshold analytic schemes through shallow learning on static feature vectors to deep sequence models ; reported accuracy has correspondingly risen from the low-97% range to above 99%. Second, despite this progress, almost all recent high-accuracy studies evaluate a standalone classifier that directly issues the trip decision, leaving the dependability–security trade-off implicit and the false-trip rate unquantified under adverse, out-of-distribution conditions. Third, none of the surveyed works couples the learned classifier to a conventional, safety-certified 87T element through an explicit supervisory handshake; consequently, a single classifier error translates directly into a maloperation. The present thesis is distinguished on all three counts: it quantifies security explicitly, it stress-tests well beyond the training distribution, and it subordinates the LSTM to the classical relay through an AND/veto gate that preserves deterministic, explainable behaviour.

It is also instructive to contrast the feature-engineering philosophy across studies. Several works pursue high-dimensional descriptors—36-component wavelet statistics , full wavelet-packet energy maps , or raw waveform windows fed to convolutional layers . Such representations can be discriminative but inflate inference cost and obscure the physical meaning of each input. The present work deliberately adopts a parsimonious, physics-informed six-feature vector (approximation and detail energy per phase), demonstrating in Chapter 6 that compact, interpretable features paired with a temporal model match or exceed the accuracy of high-dimensional alternatives while remaining deployable on standard relay hardware.

Finally, the validation modality deserves emphasis. The overwhelming majority of the surveyed studies report results purely on offline simulation datasets . Field-recorded validation and hardware-in-the-loop testing remain rare. While the present work is likewise simulation-based, it advances validation realism by deploying the trained model back into the MATLAB/Simulink electromagnetic-transient environment through ONNX, executing the complete protection pipeline—feature extraction, inference, and the hybrid decision—in closed loop with the physical plant model rather than scoring a static feature matrix offline. This closed-loop co-simulation is a meaningful step toward the hardware-in-the-loop validation outlined in Chapter 7.

CHAPTER 3 MATHEMATICAL MODELING AND THEORETICAL FRAMEWORK

3.1 Introduction

This chapter establishes the mathematical foundations for the proposed DWT-LSTM transformer differential protection scheme, covering differential relaying principles (Section 3.2), magnetizing inrush theory (Section 3.3), CT saturation modeling (Section 3.4), internal fault characteristics (Section 3.5), and the classification framework (Section 3.6).

3.2 Transformer Differential Protection Fundamentals

3.2.1 Differential Current and Restraint Current

For a two-winding transformer, the differential and restraint currents are defined as:

where $\dot{I}1$, $\dot{I}2$ are phasor currents referred to the same voltage level and N1, N2 are CT ratios. Under healthy conditions $I_d \approx 0$; a small residual arises from CT mismatch, tap-changer effects, magnetizing current, and measurement errors:

3.2.2 Operating Characteristics

The dual-slope restraint characteristic defines the operating threshold I_{op} as a function of restraint current I_r, with pickup I_{pickup}, first slope S1 (25–40%) accommodating steady-state errors, and second slope S2 (50–80%) providing additional restraint during CT saturation from through-faults. The relay trips when:

$I_d > I_{op}$

3.2.3 Pickup Current

The pickup must balance sensitivity and security, $I_{d,steady,max} < I_{pickup} < I_{d,fault,min}$, typically set to 0.2–0.5 p.u. The minimum detectable SLG fault current referred to the primary is $I_{f,min} = I_{pickup} / [1 - R_f/(Z_{s,pu} + Z_{t,pu})]$.

3.3 Magnetizing Inrush Current Phenomenon

3.3.1 Core Saturation Theory

Inrush results from transient core saturation upon transformer energization. Faraday's law gives $v(t) = N\, d\varphi/dt = NA\, dB/dt$. For sinusoidal voltage $v(t) = V_m \sin(\omega t + \alpha)$, the steady-state flux is $\varphi_{ss}(t) = -\Phi_m \cos(\omega t + \alpha)$. Including the DC transient:

At worst-case energization ($\alpha = 0$, $\Phi_r = \Phi_m$), the peak flux $\Phi_{peak} = 2\Phi_m + \Phi_r = 3\Phi_m$, exceeding B_{sat} and causing severe saturation.

3.3.2 Factors Affecting Inrush

Inrush magnitude depends on switching angle α, residual flux Φ_r (0–80% of Φ_m), source impedance Z_s, core material, and transformer rating. Typical inrush ranges from 3–15× rated current, with peak flux $\Phi_{peak}(\alpha) = \Phi_m(1 + |\cos \alpha|) + \Phi_r \cos \alpha$.

3.3.3 Harmonic Characteristics

The asymmetric inrush waveform contains both odd and even harmonics. The second harmonic dominates the even spectrum (15–60% for silicon steel; 7–12% for amorphous alloys). The key harmonic restraint ratios are $k2 = (I2/I1)\times100\%$ and $k5 = (I5/I1)\times100\%$.

3.3.4 Sympathetic Inrush

Sympathetic inrush occurs when an energizing transformer's DC inrush current, flowing through the common source impedance, drives an already-energized parallel transformer into opposite-polarity saturation, producing a slowly decaying differential current. The coupled DC circuit equations yield two exponential decay modes with combined time constants, producing sympathetic inrush that can persist for tens of seconds .

3.4 Current Transformer Saturation Model

3.4.1 CT Equivalent Circuit

The CT secondary current equals the referred primary minus the excitation current through the nonlinear magnetizing branch, $I_s = I_p' - I_e$, with $I_e = V_s/(R_m + jX_m)$ and $V_s = I_s(R_s + jX_s + Z_b)$. The nonlinear magnetization curve uses the exponential model $i_e(\lambda) = A\, e^{(B|\lambda|)}\, \mathrm{sign}(\lambda) + C\lambda$, where flux linkage $\lambda = N2\cdot\varphi = L_m(I_e)\cdot I_e$ and $v_s = d\lambda/dt$.

3.4.2 Transient DC Offset and Flux Buildup

CT saturation is caused by DC offset in fault currents, $i_p(t) = I_m[\sin(\omega t + \alpha - \varphi) - \sin(\alpha - \varphi)\, e^{(-t/\tau_f)}]$. The core flux comprises AC and DC components; saturation occurs when $\lambda > \lambda_{sat}$. The time to saturation t_{sat} decreases with higher fault current, maximum DC offset, larger remanent flux, and higher burden.

3.4.3 Impact on Differential Protection

Asymmetric CT saturation during external faults produces spurious differential current $I_d = |I_{e1} - I_{e2}| \neq 0$. When one CT remains unsaturated, the maximum spurious current is approximately $I_{d,max} \approx I_{fault} \cdot (R_{b2} + R_{s2})/(R_{b2} + R_{s2} + Z_{m2})$. The second slope S2 of the restraint characteristic provides the necessary restraint.

3.5 Internal Fault Characteristics

3.5.1 Types of Internal Faults

Internal faults include phase-to-ground (SLG, most common), phase-to-phase (LL), three-phase (LLL/LLLG), and inter-turn faults. Inter-turn faults are hardest to detect since shorted turns partially cancel the affected MMF, $I_d \approx (N_s/N) \cdot I_{load}$; for 1–5% turn faults, I_d may fall below load-current level.

3.5.2 Fault Current Analysis

Using symmetrical components, an SLG fault on phase a with impedance Z_f yields $I_{a1} = I_{a2} = I_{a0} = V_f/(Z1 + Z2 + Z0 + 3Z_f)$. For a phase-to-phase fault between phases b and c, $I_{a1} = -I_{a2} = V_f/(Z1 + Z2)$. The transient fault current contains a decaying DC component, $i_f(t) = \sqrt{2} \cdot I_{rms}[\sin(\omega t + \alpha - \varphi) - \sin(\alpha - \varphi) e^{(-t/\tau_f)}]$. Fault time constants (0.02–0.05 s) are shorter than inrush (0.1–0.5 s), and fault currents have significantly lower even-harmonic content.

3.6 Mathematical Framework for Classification

3.6.1 Feature Space Formulation

Event classification is formulated as a pattern-recognition problem. The DWT-LSTM feature vector at time step t comprises the mean, standard deviation, and energy of DWT coefficients at each level. The wavelet energy at level k is:

The optimal classifier maximizes the posterior probability (Bayes' rule), $\hat{c} = \text{argmax}_c\ P(\omega_c|x) = \text{argmax}_c\ p(x|\omega_c)\ P(\omega_c)/p(x)$. The LSTM learns this mapping through a softmax output and cross-entropy loss:

Feature separability is guided by the Fisher discriminant ratio, $FDR = (\mu_{c1} - \mu_{c2})^2 / (\sigma_{c1}^2 + \sigma_{c2}^2)$.

3.6.2 Performance Metrics

Classification performance is evaluated using confusion-matrix-derived metrics: $Accuracy = \Sigma_c TP_c / N_{total}$; $Precision_c = TP_c/(TP_c + FP_c)$; $Recall_c = TP_c/(TP_c + FN_c)$; and the F1-score $F1_c = 2 \cdot Precision_c \cdot Recall_c/(Precision_c + Recall_c)$. For protection applications, $Dependability = TP_{fault}/(TP_{fault} + FN_{fault})$ and $Security = TN_{fault}/(FP_{fault} + TN_{fault})$ are paramount. Additional metrics include the Matthews Correlation Coefficient, operating time $T_{op} = t_{trip} - t_{event}$ (within 20–40 ms at 50 Hz), and the AUC-ROC.

3.7 Adaptive Threshold and Decision-Theoretic Foundations

3.7.1 Bayes-Optimal Decision Boundary

The four-class discrimination problem can be cast in a decision-theoretic framework that makes explicit the cost asymmetry inherent to protection. Let the class-conditional densities of the feature vector x be $p(x \mid \omega_c)$ for c {internal, external, inrush, normal}, with priors $P(\omega_c)$. The Bayes risk associated with deciding class i when the true class is j is encoded by a loss matrix L_{ij}. The optimal decision minimises the conditional risk:

For protection, the loss of failing to detect an internal fault (a missed trip, FN) vastly exceeds the loss of a spurious trip (FP), which in turn exceeds the loss of an inter-class confusion among non-internal events. Encoding L_{FN} L_{FP} L_{other} biases the boundary toward dependability, mathematically formalising the engineering imperative that no internal fault may go undetected. The hybrid architecture of Chapter 5 realises this asymmetry structurally: the LSTM is permitted to veto (raise security) only after the classical 87T element has asserted a fault, so the dependability floor is set by the certified relay rather than by the learned model.

3.7.2 Adaptive Pickup as a Function of Operating State

Conventional relays employ a fixed pickup I_{pickup}. An adaptive formulation lets the threshold track the estimated steady-state error envelope so that sensitivity is maximised at light load and security is maximised during through-faults. A representative adaptive law is:

where I_{p0} is the base pickup, $I_r(t)$ the instantaneous restraint current, $\hat{\sigma}_d(t)$ an online estimate of the differential-current noise standard deviation, and k_1, k_2 tuning coefficients. The term $k_1 I_r(t)$ reproduces the classical percentage slope, while $k_2 \hat{\sigma}_d(t)$ inflates the threshold transiently when CT saturation or measurement noise elevates the differential-current variance. In the proposed scheme, this adaptive 87T element supplies the binary dependability signal, and the LSTM confidence score Conf(t) provides the complementary security signal, the two being combined in the handshake of Equation (5.1).

3.7.3 Information-Theoretic Feature Separability

Beyond the Fisher discriminant ratio of Section 3.6, the discriminative value of a feature can be quantified by the mutual information between the feature and the class label, $I(X; Y) = H(Y) - H(Y \mid X)$, where $H(\cdot)$ denotes Shannon entropy. A feature with high $I(X; Y)$ reduces class uncertainty substantially and is therefore retained. The wavelet detail energies are expected to exhibit the highest mutual information because fault inception injects broadband high-frequency content that the detail sub-bands localise efficiently, whereas inrush concentrates energy near the second harmonic and steady-state operation produces negligible detail energy. This expectation is confirmed empirically in Chapter 6 (Section 6.2.3), where the three detail-energy features account for 72.3% of the total mutual information.

3.7.4 Operating-Time Bound

The maximum permissible operating time T_{max} is dictated by transformer through-fault withstand and downstream coordination. For a 50 Hz system, sub-cycle clearing ($T_{op} < 20$ ms) is the design target; the thermal stress imposed on the winding during an internal fault scales with the integral $\int i^2(t)\,dt$ over the fault duration, so every millisecond of reduced clearing time directly lowers insulation degradation and mechanical force on the windings. The latency budget derived in Chapter 5 (Section 5.6.3) and validated in Chapter 6 (Section 6.4.3) demonstrates that the complete DWT–LSTM–hybrid pipeline satisfies this bound with substantial margin.

CHAPTER 4 POWER SYSTEM SIMULATION MODEL USING MATLAB/SIMULINK

4.1 Introduction

MATLAB/Simulink was selected as the primary simulation platform for this research due to its uniquely integrated environment that combines high-fidelity electromagnetic-transient modelling via Simscape Electrical with native machine-learning support through the Deep Learning Toolbox and ONNX runtime. This integration enables a seamless workflow from simulation-based data generation to DWT-LSTM model training, validation, and deployment within a single ecosystem. Simscape Electrical provides validated component models—saturable transformers, CTs with core saturation, circuit breakers, and programmable voltage sources—ensuring waveform-level fidelity essential for protection studies, and supports automated batch simulations via MATLAB scripting .

4.2 MATLAB/Simulink Simulation Environment

4.2.1 Software Overview

The simulation platform comprises MATLAB R2024a with Simulink, Simscape Electrical, the Deep Learning Toolbox, the Signal Processing Toolbox, and the ONNX converter for MATLAB. Simscape Electrical uses a physical-network approach where components enforce Kirchhoff's laws, with the underlying solver automatically constructing and solving network equations at each time step. The Deep Learning Toolbox supports import and inference of the trained DWT-LSTM model via ONNX .

4.2.2 Solver Configuration

A discrete-time solver with a fixed sample time of $\Delta t = 1\times10^{-5}$ s (10 μs) is employed, corresponding to a Nyquist frequency of $f_{Nyquist} = 1/(2\Delta t) = 50$ kHz. This provides a 50× oversampling margin relative to the relay rate of 1600 Hz (32 samples/cycle at 50 Hz), ensuring negligible aliasing. The powergui block uses Discrete mode at 10 μs .

4.2.3 Simulation Duration

Each simulation runs for T_{sim} = 1.0 s, with the event injected at t_{event} = 0.2 s (10 cycles pre-event, 40 cycles post-event). At the relay sampling rate of 1600 Hz, $N_{samples} = 1600 \times 1.0 = 1600$; including the initial reference, each record contains 1601 samples (n = 0 to 1600), yielding $1601 - 32 + 1 = 1570$ one-cycle sliding windows per case.

4.3 Transformer Model Parameters

4.3.1 Transformer Rating and Configuration

The protected transformer is a three-phase, 300 MVA, 230/11 kV unit with Yd1 vector group (grounded wye primary, delta secondary with −30° phase displacement), a prevalent configuration in transmission substations . The full-load currents are $I_{FLA,HV} = 300\times10^6/(\sqrt{3}\cdot230\times10^3) = 753.1$ A and $I_{FLA,LV} = 300\times10^6/(\sqrt{3}\cdot11\times10^3) = 15\ 746$ A. Per-unit base impedances are $Z_{base,HV}$ = 176.33 Ω and $Z_{base,LV}$ = 0.4033 Ω. Parameters are summarized in Table 4.1.

4.3.2 Core Saturation Characteristics

The Simscape Electrical saturable-transformer block models core nonlinearity through a piecewise-linear B-H curve specified as flux-linkage / magnetizing-current coordinate pairs, configured with 20 data points spanning ±2.0 p.u. flux and a knee point near 1.2 p.u. flux. Above the knee, incremental permeability drops to 0.2% of the air-core inductance, reproducing the deep-saturation characteristic of grain-oriented silicon steel (M5 grade) . Worst-case inrush ($\alpha = 0°$, $\varphi_r = +\varphi_m$) yields $\varphi_{peak} = 3\varphi_m$, producing inrush peaks of 5–10 times rated current .

4.3.3 Three-Phase Winding Configuration

The Yd1 group is implemented with the primary as “Yg” and secondary as “D1” (delta, −30° lag). The 30° displacement requires phase compensation before computing the differential current, performed by applying a −30° rotation to the LV-side CT currents via a Clarke transformation.

4.4 Power System Network Model

4.4.1 Source Model and Equivalent Impedance

The external system is represented by Thévenin equivalents on both sides . On the HV side (1.05 p.u. voltage, X/R = 30.0), $Z_{th,HV} = 0.5 + j15.0$ Ω with a short-circuit level of 3 524 MVA. On the LV side (1.0 p.u. voltage), $Z_{th,LV} = 0.01 + j0.10$ Ω with 1 204 MVA. Available fault currents are $I_{f,HV}$ = 8.84 kA and $I_{f,LV}$ = 63.2 kA, consistent with typical 230/11 kV substations.

4.4.2 Breaker and Snubber Circuit

Circuit breakers are modelled using the Simscape Electrical three-phase breaker block (R_on = 0.001 Ω, R_off = 10^6 Ω). RC snubber circuits (R_s = 1000 Ω, C_s = 0.1 μF) are connected across each breaker to damp high-frequency switching transients and maintain numerical stability of the discrete solver .

4.4.3 Load Model

The LV-side load is a composite of 70% constant impedance (168 MVA) and 30% constant power (72 MVA), totalling 240 MVA at 0.85 lagging power factor (80% transformer loading). A 2% phase unbalance and a shunt capacitor bank for power-factor correction are included.

4.4.4 Current Transformer Model

CTs on both sides are modelled using the Simscape Electrical linear-transformer block with saturable core. Accurate CT modelling is critical since CT saturation during through-faults is a well-documented cause of differential-relay maloperation . CT saturation is modelled through a piecewise-linear flux-current curve. Parameters are given in Table 4.2.

4.5 Fault and Event Simulation Scenarios

4.5.1 Internal Fault Modeling

Internal faults are simulated at five winding taps (5%, 25%, 50%, 75%, 95%) using the three-phase fault block. Four fault types (SLG, LL, LLG, 3LG) are applied with three inception angles and three fault resistances (0.01 Ω, 5 Ω, 25 Ω). The effective fault impedance at tap p is $Z\_fault(p) = p^2 Z\_k + R\_f$. Terminal faults yield the highest currents; 720 deterministic cases are generated .

4.5.2 External Fault Modeling

External faults (SLG, LL, LLG, 3LG) are simulated on both HV and LV buses, representing faults outside the differential zone. Evolving faults (SLG → LLG after 3 cycles → 3LG after 6 cycles) are implemented via a Simulink state machine. Stress-test scenarios include asymmetrical CT saturation and fault currents ranging from 5 to 30 p.u., producing 540 deterministic external-fault cases .

4.5.3 Inrush Current Simulation

Inrush conditions are generated by sweeping the switching angle α across seven values (0° to 180° in 30° steps) and the residual flux φ_r across nine values (−0.8 to +0.8 p.u.), yielding 63 combinations. Simulations include symmetric, asymmetric, and single-phase residual-flux distributions. Sympathetic inrush is modelled using two parallel transformers, producing 189 deterministic inrush base cases .

4.5.4 CT Saturation Conditions

CT saturation stress cases vary burden impedance (rated, 2×, 4×), fault-current magnitude (10, 20, 30× rated), and remanent flux (0, $+0.5B_r$, $+0.8B_r$), producing 108 dedicated cases spanning both bus sides and worst-case inception angles . These are grouped with external faults for classification (incl. CT saturation).

4.5.5 Normal Operating Conditions

Normal operating scenarios cover six load levels (0% to 110% rated), three tap positions, five voltage levels (0.95–1.05 p.u.), and three power factors. The steady-state differential current due to tap mismatch is $I_{diff,tap} = I_{load} \cdot \Delta_{tap}/(1 + \Delta_{tap})$, yielding approximately 5% of load current for a ±5% tap change. A total of 270 deterministic normal-operating cases are generated .

4.6 Automated Data Generation and Dataset Summary

The batch simulation is automated via a MATLAB script (ControlPanel.m) that modifies Simulink parameters via set_param, runs simulations in parallel using parsim, and post-processes results. The signal-conditioning pipeline emulates a merging unit: a 4th-order Butterworth anti-aliasing filter (800 Hz cut-off), 16-bit quantization, and AWGN at SNR levels of 20, 30, and 40 dB. After downsampling to 1600 Hz, each case yields a 1601-sample record per phase, exported to .mat files for the PyTorch training pipeline. Stochastic augmentation randomizes inception angle U(0°, 360°), fault resistance logU(0.01, 50) Ω, source impedance (±5%), and fault location U(2%, 98%), expanding and balancing the deterministic base cases to a final dataset of 12,800 cases (3,200 per class) .

4.7 Signal Conditioning and Merging-Unit Emulation

To ensure that the generated dataset reflects the measurement chain of a modern digital substation rather than idealised simulator output, each raw differential-current record is passed through a signal-conditioning pipeline that emulates an IEC 61850-9-2 merging unit. The pipeline comprises four stages applied in sequence: anti-alias filtering, decimation to the relay rate, quantisation, and noise injection. Table 4.4 lists the parameters of each stage.

The 800 Hz anti-alias cut-off is matched to the 1.6 kHz Nyquist limit, attenuating components above the eighth harmonic that would otherwise alias into the DWT sub-bands. Sixteen-bit quantisation reproduces the dynamic range of contemporary analogue-to-digital front ends, introducing a quantisation floor well below the noise levels of interest. The additive white Gaussian noise model conservatively represents combined sensor, electromagnetic-interference, and transmission noise; the three nominal SNR levels populate the training set, while the 15 dB and 50 dB extremes are reserved for the out-of-distribution stress tests of Chapter 6. This conditioning chain is applied identically during data generation and during Simulink deployment, so that no distribution shift is introduced between training and inference.

4.8 Dataset Statistics and Class Balance

The final balanced corpus comprises 3,200 cases per class (12,800 total), with stratified 70/15/15 partitioning preserving class proportions in every subset. Balancing is achieved by combining deterministic base scenarios (Section 4.5) with stochastic augmentation that randomises inception angle, fault resistance, source impedance, and fault location, ensuring that each class spans a comparable region of the operating-condition space. Figure 4.3 visualises the composition of the dataset across classes and subsets, and the parameter ranges spanned by the augmentation are summarised in Table 4.5.

CHAPTER 5 PROPOSED DWT-LSTM METHODOLOGY

5.1 Introduction

Conventional harmonic-restraint differential protection for power transformers suffers from well-documented limitations: second-harmonic content in modern high-permeability transformers can fall below typical 15–20% restraint thresholds during inrush, internal faults with CT saturation generate even-harmonic content that causes failure to trip, and power electronic devices alter harmonic signatures unpredictably. This chapter presents the proposed hybrid adaptive DWT-LSTM framework that addresses these limitations through a two-step architecture combining MATLAB/Simulink for physical power-system modelling and real-time deployment with Python/PyTorch for deep-learning model training. The framework decomposes differential current signals using a three-level Daubechies-4 (db4) DWT and extracts two energy features per phase based on Parseval's theorem, yielding a compact 6-dimensional feature vector accumulated over a 32-sample full-cycle buffer. A two-layer LSTM network with global temporal attention processes the resulting tensor to classify events into four categories: internal fault, external fault, inrush current, and normal condition. Critically, the LSTM operates as a supervisory veto gate within a hybrid decision engine: a trip command is issued only when both the classical adaptive 87T element operates and the LSTM classifies the event as an internal fault with confidence exceeding a predefined threshold.

5.2 Framework Architecture Overview

5.2.1 Two-Step Architectural Approach

Step 1 — MATLAB/Simulink Physical Modelling and Data Generation. A detailed electromagnetic-transient model is constructed in MATLAB/Simulink using the Simscape Electrical toolbox, including the three-phase power transformer (with nonlinear saturation and remanent flux), CTs on both HV and LV sides (with saturation and remanence), the transmission network, and configurable fault modules. The differential current for each phase is computed at a native relay sampling rate of 1.6 kHz (32 samples per 50 Hz cycle), yielding 1601 discrete time steps for a 1.0-second simulation. Three-phase differential current waveforms with event labels are exported to .mat files in MATLAB's column-major layout.

Step 2 — Python/PyTorch AI Training and ONNX Export. The .mat files are loaded via scipy.io.loadmat, converted from column-major to row-major layout for PyTorch, and processed by a DWT feature-extraction engine (PyWavelets) that computes approximation and detail energy features from 16-sample half-cycle windows. The resulting tensors of shape feed into a PyTorch LSTM model with global temporal attention, trained using the Adam optimizer with weighted cross-entropy loss. The trained model is exported to ONNX at Opset 14 via torch.onnx.export, producing a stateless inference graph.

Step 2b — Simulink Deployment. The ONNX model is loaded into Simulink via the Deep Learning Toolbox Predict block, which performs real-time inference at 1.6 kHz. The Predict block receives the 6-dimensional energy feature vector at each time step, buffers it internally into the tensor, and outputs a confidence score and class prediction that feed the hybrid decision engine alongside the classical adaptive 87T element. The two-step approach enables independent development and validation of the power-system model and AI model, GPU-accelerated training in Python, vendor-neutral ONNX portability, and native closed-loop HIL testing via Simulink.

5.2.2 The Hybrid Decision Handshake

Neural-network-based approaches for transformer differential protection have been explored extensively , with recent work on deep-learning integration . The proposed LSTM classifier operates as a supervisory veto gate rather than a replacement for the classical relay, motivated by the requirement for deterministic, explainable relay behaviour. The hybrid decision handshake implements a logical AND condition:

TRIP = Adaptive_87T ($\hat{c}$ = Internal Fault) (Conf ≥ θ_conf)

where Adaptive_87T is the binary output of the classical adaptive differential relay element, $\hat{c}$ is the LSTM-predicted class label, and Conf is the LSTM confidence score. The trip command is issued only when all three conditions are simultaneously satisfied. This provides: (1) dependability preservation—the classical 87T element retains primary fault-detection responsibility; (2) security enhancement—the LSTM veto blocks false trips during inrush, CT saturation, and external faults; and (3) fail-safe operation—on LSTM inference failure, the logic defaults to the classical 87T output. The confidence threshold θ_conf further blocks trips during ambiguous events.

5.2.3 Block Diagram Description

The complete signal flow from raw CT secondary currents to the final trip/block command comprises seven sequential stages. Stage 1 — CT Secondary and Merging Unit: three-phase secondary currents are digitised at f_s = 1.6 kHz via IEC 61850-9-2 SV, and the differential current for each phase is computed as $i\_diff,\varphi(n) = i\_HV,\varphi(n) + i'\_LV,\varphi(n)$, where $i'\_LV,\varphi(n)$ includes CT ratio correction and Yd1 phase-shift compensation (−30°). Stage 2 — Half-Cycle Sliding Window: each phase is buffered into a 16-sample window (10 ms) updated every 0.625 ms. Stage 3 — DWT Engine: each window undergoes 3-level db4 decomposition, yielding a 6-element energy feature vector. Stage 4 — Full-Cycle Buffer: features are accumulated over 32 samples and reshaped to . Stage 5 — LSTM Classification: the ONNX Predict block produces the softmax probability vector and confidence score Conf = max(ŷ). Stage 6 — Hybrid Decision Engine: the AND condition of Equation (5.1) determines the trip/block command. Stage 7 — Circuit Breaker Trip Coil: the trip command is latched and sent to the breaker with a minimum duration of 40–60 ms.

5.3 Data Preprocessing and Digital Sampling

5.3.1 1.6 kHz Discrete Sampling Rate

The framework adopts $f\_s = 1.6$ kHz, providing 32 samples per 50 Hz cycle, $f\_s = N \times f\_0 = 32 \times 50 = 1600$ Hz, with inter-sample spacing $\Delta t = 1/f\_s = 0.625$ ms. This rate satisfies the Nyquist criterion for components up to the 10th harmonic (500 Hz), provides a Nyquist frequency of 800 Hz, and uses a power-of-two sample count enabling clean dyadic DWT decomposition of the 16-sample half-cycle window (16→8→4→2 coefficients) without zero-padding. The 1.6 kHz rate also aligns with IEC 61850-9-2 merging-unit capabilities. The relay/feature-extraction subsystem operates at a sample time of $T\_s = 6.25\times10^{-4}$ s (1.6 kHz), producing 1601 samples per 1.0-second simulation, while the underlying Simscape electromagnetic-transient solver runs at 10 µs (Section 4.2.2).

5.3.2 Sliding Window Implementation

The framework employs two sliding windows at different time scales. The half-cycle window for DWT (16 samples, 10 ms) buffers the most recent differential current samples for each phase, shifting forward by one sample at each 1.6 kHz acquisition. The 16-sample length matches the dyadic requirements of the 3-level DWT, producing 2 coefficients at the deepest level—the minimum for meaningful energy computation. The full-cycle buffer for the LSTM (32 samples, 20 ms) accumulates the 6-element feature vectors into a 32×6 matrix, reshaped to before inference. This decouples the feature-extraction time scale (half-cycle, for fast response to transients) from the classification time scale (full-cycle, for temporal pattern discrimination). In Simulink, both buffers use MATLAB Function blocks with persistent circular buffers, and the reshape uses Permute Dimensions and Reshape blocks.

5.3.3 Column-Major to Row-Major Data Conversion

MATLAB stores arrays in column-major (Fortran-style) order while PyTorch uses row-major (C-style) order. The .mat file stores data as 1601×3 (time steps × phases) in column-major order. After loading via scipy.io.loadmat, an explicit transpose produces [N_scenarios, 3, 1601] tensors for PyTorch, with np.ascontiguousarray ensuring physical memory rearrangement for optimal GPU access. The complete data pipeline is summarised in Table 5.1.

5.4 DWT Engine and Feature Extraction

5.4.1 Daubechies-4 (db4) Mother Wavelet Selection

The db4 wavelet is selected for its compact support (L = 8 coefficients, 4.375 ms at 1.6 kHz), 4 vanishing moments, orthogonality, and morphological similarity to fault inception transients. Its jagged, asymmetrical shape closely matches the abrupt, non-sinusoidal wavefronts of fault-induced discontinuities, enabling efficient energy capture with minimal coefficient spread—unlike smoother wavelets (e.g., Morlet) that disperse energy across levels. The 8-coefficient filter length provides sharp temporal localization within the 16-sample half-cycle window while maintaining adequate frequency selectivity.

5.4.2 Three-Level Wavelet Decomposition

The 3-level DWT of the 16-sample half-cycle window partitions the spectrum into four sub-bands. The decomposition follows the Mallat algorithm , applying successive convolution with quadrature mirror filters h[n] (low-pass) and $g[n] = (-1)^n h[L-1-n]$ (high-pass, L = 8), followed by down-sampling by 2:

where a_j and d_j are the approximation and detail coefficients at level j, with a_0 = x(t) being the original 16-sample input. The frequency band allocation is shown in Table 5.2.

The dyadic structure yields N_j = 16/2^j coefficients per level: d1 has 8, d2 has 4, and a3, d3 each have 2, totalling 16 (equal to the input length, confirming orthogonality). The 3-level depth is the maximum useful decomposition for a 16-sample window—level 4 would yield single coefficients, providing no meaningful energy information—while covering 0–800 Hz with efficient arithmetic cost.

5.4.3 Approximation and Detail Energy Calculation

Two energy features are extracted per phase based on Parseval’s theorem, which guarantees energy conservation in the wavelet domain. The approximation energy captures low-frequency content (0–100 Hz):

E_A increases sharply during internal faults (large fundamental current) and exhibits a characteristic decaying pattern during inrush. The detail energy captures high-frequency content (100–800 Hz):

E_D rises sharply at fault inception (high-frequency transients) while inrush produces moderate values dominated by second-harmonic content . The implicit E_A/E_D ratio provides an additional discriminant that the LSTM can learn through its gated memory architecture.

5.4.4 Feature Vector Construction

The 6-dimensional feature vector concatenates energy features from all three phases, $f(t) = [E_{A,A}, E_{D,A}, E_{A,B}, E_{D,B}, E_{A,C}, E_{D,C}]$. This compact representation minimizes inference cost, provides physics-informed spectral energy information, and preserves per-phase discrimination essential for detecting single-phase and phase-to-phase faults. Features are normalised using z-score normalisation, $f(t) = (f(t) - \mu)/\sigma$, where μ and σ are the training-set mean and standard deviation, applied consistently during training and Simulink deployment (implemented as Gain and Bias blocks). The normalised features are accumulated over the full-cycle buffer into $X(t)$ $\mathbb{R}^{32\times 6}$, reshaped to the final input tensor X_{LSTM} $\mathbb{R}^{1\times 32\times 6}$.

5.5 LSTM Architecture and Training Pipeline

5.5.1 Model Architecture

The LSTM network processes input tensors of shape [B, 32, 6] and produces a 4-class probability output with an associated confidence score. The architecture comprises: (1) an input layer accepting [B, 32, 6]; (2) a first LSTM layer with 128 hidden units (return_sequences = True) followed by dropout ($p = 0.3$); (3) a second LSTM layer with 64 hidden units (return_sequences = True) followed by dropout ($p = 0.3$); (4) a global temporal attention layer computing a weighted sum across the 32 time steps:

where h_t $\mathbb{R}^{64}$ is the second LSTM layer's hidden state, and W_a, b_a, v are learned attention parameters. This mechanism enables the network to focus on the most informative time steps (e.g., post-fault inception) while providing interpretability through the learned attention weights; (5) a dense layer with 32 units and ReLU activation; and (6) an output layer with 4 units and softmax activation, producing $\hat{y} = [\hat{p}_{internal}, \hat{p}_{external}, \hat{p}_{inrush}, \hat{p}_{normal}]$. The LSTM cell computes its internal state using the standard gating equations (forget, input, candidate, cell, output gates) defined in Section 2.5.1. The total trainable parameters are $P_{total} = 125{,}476$, comprising 69,632 (LSTM 1), 49,408 (LSTM 2), 4,224 (attention), 2,080 (dense), and 132 (output).

5.5.2 Training Procedure

The model is trained in PyTorch using weighted categorical cross-entropy loss to handle class imbalance, $L(y, \hat{y}) = -\Sigma_c\, w_c\, y_c \log(\hat{p}_c)$, with class weights $w_c = N_{total}/(C \cdot N_c)$, where $C = 4$ and N_c is the count for class c. The Adam optimiser ($\eta = 0.001$, $\beta 1 = 0.9$, $\beta 2 = 0.999$) is used with a ReduceLROnPlateau scheduler (factor 0.5, patience 10 epochs, minimum 10^{-6}). Batch size is 128 with a maximum of 200 epochs, and early stopping monitors validation loss with patience of 20 epochs. The confidence score for the hybrid decision engine is $Conf(t) = \max_c \hat{p}_c(t)$, with $\theta_{conf} = 0.85$ determined from validation-set analysis.

5.5.3 Hyperparameter Optimization and Cross-Validation

A stratified 5-fold cross-validation strategy evaluates generalisation performance, reporting mean and standard deviation of accuracy, precision, recall, and F1-score across folds. Hyperparameters (hidden units, dropout rate, learning rate, decomposition level, mother wavelet) are optimised via grid search. Stress testing includes extreme fault resistances (0.01–300 Ω), simultaneous events, evolving faults, and near-boundary conditions (low-magnitude internal faults, inrush with borderline second-harmonic content).

5.5.4 ONNX Export

The trained PyTorch model is exported using torch.onnx.export with Opset 14 (broad operator coverage compatible with the MATLAB Deep Learning Toolbox), a fixed input shape for deterministic inference, and a stateless design in which the full 32-step sequence is provided at each call with no hidden-state carryover. The resulting .onnx file is loaded in Simulink via importNetworkFromONNX for integration with the physical power-system model.

5.6 Real-Time Implementation in Simulink

5.6.1 Predict Block Integration

The ONNX model is integrated via the Deep Learning Toolbox Predict block configured for the dwt_lstm_relay.onnx file with softmax probability output. The Predict block is encapsulated in a masked Simulink subsystem ("DWT-LSTM Classifier") along with the feature-extraction, normalisation, and tensor-assembly stages. Internally, a MATLAB Function block maintains a circular buffer of the 32 most recent 1×6 feature vectors, with Permute Dimensions and Reshape blocks converting to format. This modular design allows ONNX model replacement without modifying the surrounding protection logic.

5.6.2 Hybrid Decision Engine

The hybrid decision engine is implemented using Simulink logic blocks and a Stateflow chart encoding the trip/block state machine. Three binary conditions are evaluated: Adaptive_87T (classical element operated), LSTM_Internal ($\hat{c}$ = internal), and High_Confidence (Conf $\geq$ 0.85), combined via TRIP = Adaptive_87T LSTM_Internal High_Confidence. The Stateflow chart latches the trip output for 100 ms once asserted. The LSTM veto blocks false trips when the classical 87T operates but the classifier disagrees; conversely, the classical element prevents the LSTM from initiating trips independently, maintaining the dependability guarantee.

5.6.3 Computational Latency

The total end-to-end processing latency is estimated as $T_{proc} = T_{buffer} + T_{DWT} + T_{LSTM} + T_{logic} \approx 5.0 + 0.1 + 1.0 + 0.01 = 6.1$ ms, corresponding to approximately 0.3 power-frequency cycles, well within the one-cycle (20 ms) requirement. The dominant component is the quarter-cycle buffer fill, which is pipelined after initialisation. LSTM ONNX inference contributes 0.5–2.0 ms on CPU (<0.1 ms on GPU). The total computational cost is approximately 1.31 MFLOPs per time step, translating to ~0.66 ms on a 1 GHz DSP with SIMD, confirming real-time feasibility on standard protection hardware.

5.7 Robustness Enhancement

The framework employs three complementary robustness strategies. Noise augmentation: Additive White Gaussian Noise (AWGN) is injected into CT secondary currents during MATLAB simulation at SNR levels of 20, 30, and 40 dB, with additional stress testing at 15 dB and 50 dB. The DWT energy features exhibit inherent noise resilience due to the frequency-localised decomposition acting as an energy-domain low-pass filter. Dropout regularisation: A dropout rate of 0.3 between LSTM layers prevents co-adaptation of hidden units and improves generalisation. Out-of-distribution stress testing evaluates the model under switching angles (0–360°), frequency deviations (47–53 Hz), remanent flux mismatches (±0.9 B_{sat}), and CT ratio errors (±5%), with all scenarios processed end-to-end through the Simulink-embedded ONNX model to validate the complete pipeline beyond the training distribution.

5.8 Chapter Summary

This chapter presented the complete methodology of the proposed hybrid adaptive DWT-LSTM framework. The key contributions include: (1) a two-step architecture decoupling MATLAB/Simulink physical modelling from Python/PyTorch AI training, connected via ONNX export at Opset 14; (2) a 1.6 kHz sampling rate with 32 samples per cycle providing dyadic-compatible DWT decomposition; (3) a 3-level db4 DWT on 16-sample half-cycle windows producing two energy features (E_A, E_D) per phase based on Parseval's theorem; (4) a compact 6-dimensional feature vector accumulated into LSTM input tensors; (5) a two-layer LSTM (128–64 hidden units) with global temporal attention and 125,476 parameters; (6) a hybrid decision engine implementing AND logic between the classical adaptive 87T element and the LSTM classifier with confidence thresholding; and (7) a demonstrated total processing latency of approximately 6.1 ms (0.3 cycles), confirming real-time feasibility.

5.9 Adaptive Decision Logic and Confidence Calibration

While the core handshake of Equation (5.1) uses a fixed confidence threshold $\theta_{conf} = 0.85$, the framework supports a state-dependent threshold that tightens or relaxes the security requirement according to the prevailing operating context. The adaptive confidence threshold is defined as:

where $\theta_0 = 0.85$ is the nominal threshold, the indicator $1[I_r > I_{r,hi}]$ lowers the requirement during high-restraint conditions (genuine high-current internal faults, where dependability is paramount), and the indicator $1[K_s > K_{s,hi}]$ raises the requirement when a CT-saturation index K_s is high (where false trips are most likely). The coefficients γ_1, γ_2 are tuned on the validation set. This mechanism allows the relay to remain maximally dependable for severe internal faults while becoming maximally secure precisely when external disturbances threaten to provoke a spurious trip.

Because neural-network softmax outputs are frequently over-confident, the raw probabilities are temperature-calibrated before thresholding. Given logits z, the calibrated probability is $\hat{p}_c = \mathrm{softmax}(z/T)_c$, where the scalar temperature $T > 1$ is fitted by minimising the negative log-likelihood on the validation set. Calibration ensures that a reported confidence of 0.85 corresponds to an empirical correctness probability near 0.85, so that θ_{conf} carries a meaningful, well-defined operating point on the reliability diagram.

5.9.1 Probabilistic Uncertainty Layer

To distinguish confident misclassifications from genuinely ambiguous events, the framework exposes the full softmax distribution rather than only the arg-max. The predictive entropy $H(\hat{y}) = -\Sigma_c \hat{p}_c \log \hat{p}_c$ quantifies decision uncertainty: low entropy with high max-probability indicates a confident, trustworthy decision; high entropy flags an ambiguous event for which the hybrid logic conservatively withholds the trip and, optionally, raises a supervisory alarm. This uncertainty-aware behaviour is particularly valuable for borderline cases such as low-magnitude internal faults and inrush events with marginal second-harmonic content.

5.10 Computational Complexity Derivation

The per-time-step computational cost is dominated by the LSTM forward pass. For an LSTM layer with input dimension d_{in} and hidden dimension d_h, each gate requires $4\cdot(d_{in}\cdot d_h + d_h^2)$ multiply-accumulate operations per time step. For the first layer ($d_{in} = 6$, $d_h = 128$) and second layer ($d_{in} = 128$, $d_h = 64$), summed over the 32-step sequence, the total approaches 1.3 MFLOPs per inference, consistent with the figure reported in Section 5.6.3. The DWT front end adds only $O(N\cdot L)$ operations for an N-sample window and an L-tap filter ($N = 16$, $L = 8$), a negligible fraction of the total. The attention layer contributes $O(32\cdot d_h)$ operations. Table 5.3 itemises the parameter and operation budget per stage.

This budget translates to approximately 0.66 ms on a 1 GHz DSP with single-instruction-multiple-data (SIMD) acceleration, or below 0.1 ms on a modern GPU, confirming that the model is light enough for embedded deployment within a numerical relay. The compactness is a direct consequence of the parsimonious six-feature representation: feeding raw 32-sample, three-phase windows to the network instead would inflate the first-layer input dimension and the corresponding operation count by roughly an order of magnitude.

CHAPTER 6 RESULTS, ANALYSIS AND DISCUSSION

6.1 Introduction

This chapter presents a comprehensive evaluation of the proposed Hybrid Adaptive Transformer Differential Protection (HATDP) framework, in which a Discrete Wavelet Transform–Long Short-Term Memory (DWT-LSTM) classifier serves as a supervisory layer over a conventional 87T differential relay. All simulation data were generated in the MATLAB/Simulink environment for a 300 MVA, 230 kV/11 kV, Yd1 power transformer with three-phase differential currents sampled at 1.6 kHz (32 samples/cycle). A sliding-window db4 DWT was applied to each half-cycle window, producing six energy features per classification instant. A comprehensive dataset of 12,800 event cases was generated covering normal load, external faults, magnetizing inrush, and internal faults, partitioned into training (70%), validation (15%), and testing (15%) subsets—yielding a 1,920-case test set (480 cases per class). The assessment spans feature-extraction quality, offline classification accuracy, real-time ONNX inference latency, hybrid veto/AND-gate behaviour, stress testing under adverse conditions, and benchmarking against existing methods.

6.2 Data Generation and Preprocessing Results

6.2.1 MATLAB/Simulink Simulation Data Quality

6.2.2 DWT Feature Distribution Analysis

Six energy features are computed per classification instant: approximation energy E_A (0–100 Hz) and detail energy E_D (100–800 Hz) for each of the three phases. Table 6.1 presents the statistical distribution across event classes from the 1,920-case test set. The detail-energy features provide the primary discriminant, with Phase C E_D showing an inter-class separation ratio of 502× between internal faults and normal operation. Approximation energy distinguishes inrush ($\approx 1.8\times10^{-3}$) from external faults ($\approx 0.4\times10^{-3}$). Three-phase symmetry is maintained within 12% inter-phase variation .

6.2.3 Feature Importance Analysis

Mutual information (MI) analysis using the adaptive k-NN estimator ($k = 5$) quantifies the relative contribution of each feature (Table 6.2). Detail-energy features collectively account for 72.3% of the total MI. An ablation study confirms that detail-only features achieve 98.83% accuracy, approximation-only degrades to 91.24%, and the full six-feature set achieves 99.27%.

6.3 LSTM Classification Performance (Offline Training)

6.3.1 Training and Validation Curves

The LSTM model (two layers of 128 and 64 hidden units, dropout 0.3) was trained in PyTorch with the Adam optimizer ($\alpha = 0.001$) and weighted categorical cross-entropy loss. Training loss converged from 1.381 to 0.0142 at epoch 94 (early stopping), with validation loss at 0.0198 ($\Delta = 0.0056$, indicating minimal overfitting). Validation accuracy reached 99.48%, with 95% achieved within 18 epochs.

6.3.2 Confusion Matrix Analysis

6.3.3 Per-Class Performance Metrics

6.3.4 K-Fold Cross-Validation Results

A 5-fold stratified cross-validation (Table 6.5) demonstrates consistent performance with mean accuracy of 99.31±0.18%. Internal fault recall remains at 100.00±0.00% across all folds—no internal fault was misclassified in any fold—providing strong evidence that the 100% dependability requirement is achieved.

6.4 Real-Time Hybrid Relay Validation (Simulink)

The trained LSTM model was exported to ONNX format via torch.onnx.export (Opset 14) and imported into Simulink using the Deep Learning Toolbox Predict block, enabling co-simulation of the physical power-system model and the AI-based supervisory protection logic.

6.4.1 ONNX Integration and Inference Latency

6.4.2 Hybrid Decision Logic Performance

The hybrid relay issues a trip only when both the 87T relay and the LSTM indicate an internal fault (AND gate). If the 87T operates but the LSTM classifies non-internal, the LSTM vetoes the trip. Table 6.7 shows the hybrid achieves 100% dependability and 100% security—zero false negatives and zero false trips—on the 1,920-case test set. The LSTM vetoes all 77 false trips from the standalone 87T relay .

6.4.3 System Clearing Time Benchmarks

The total clearing time from fault inception to trip output is less than 13.1 ms (Table 6.8), firmly sub-cycle at 50 Hz (20 ms). This represents approximately a 2× speedup over conventional harmonic-restraint relays (30–40 ms) , directly reducing transformer thermal stress (I^2t) during internal faults.

6.5 Stress Testing and Out-of-Distribution Robustness

A series of stress tests evaluates classifier resilience to parameter variations outside or at the boundary of the training distribution.

6.5.1 Varying Point-on-Wave Switching

Classification accuracy was evaluated at intermediate switching angles (15°, 45°, 75°) not present in the training data. Table 6.9 shows a minimum accuracy of 98.75% at 45° with 100% internal-fault recall maintained across all angles.

6.5.2 CT Remanent Flux Mismatch

The framework was tested with extreme remanent flux levels (±85% to ±95%) beyond the training range (±80%). Table 6.10 shows graceful degradation with 100% internal-fault recall maintained at all levels.

6.5.3 Frequency Deviation Tests

6.5.4 Measurement Noise

AWGN was injected at various SNR levels (Table 6.12). The framework maintains 98.65% accuracy at 30 dB SNR and 97.08% at 20 dB, with 100% internal recall down to 30 dB; the DWT's half-cycle integration provides inherent noise filtering.

6.5.5 High-Impedance Faults

Fault resistance was varied from 0 to 500 Ω (Table 6.13). The LSTM detects 100% of faults up to 300 Ω and 98.33% up to 500 Ω, dramatically exceeding the standalone 87T (70.83% at 500 Ω). In the hybrid architecture, cases where only the LSTM detects a fault generate a supervisory alarm, achieving 100% coverage (trip or alarm) at all resistances.

6.6 Comparative Benchmarking

6.7 Discussion of Results

The experimental results demonstrate that the proposed hybrid DWT-LSTM + adaptive 87T relay architecture achieves the critical protection requirements of zero false negatives (100% dependability), zero false trips (100% security), and sub-cycle clearing (13.1 ms) on the 12,800-case simulation dataset. The compact six-feature DWT energy representation provides robust discrimination, with detail-energy features serving as the primary fault signature and approximation-energy features distinguishing inrush from external faults. Real-time Simulink validation confirms ONNX integration feasibility with sub-millisecond inference latency, and stress testing demonstrates graceful degradation under CT saturation, frequency deviation, measurement noise, remanent flux mismatch, varying fault inception angles, and high-impedance faults—with zero false negatives maintained in all scenarios.

Several limitations must be acknowledged. The LSTM model was trained exclusively on data from a 300 MVA, 230/11 kV, Yd1 transformer; deployment on different ratings or vector groups would require retraining. The dataset does not include inter-turn winding faults, and all validation is simulation-based—field deployment would require compliance with IEC 60255, IEEE C37.91, and

IEC 62351 cybersecurity standards. Despite these limitations, the hybrid architecture's complementary design resolves the fundamental dependability–security trade-off that limits existing methods, providing a practical pathway for next-generation transformer differential protection.

6.8 Detailed Comparison with Prior Published Work

Section 6.6 benchmarked the proposed framework against re-implemented baseline algorithms on the unified 12,800-case dataset. This section broadens the comparison to the specific studies in the reference list, examining not only headline accuracy but the qualitative capabilities that determine field viability: explicit temporal modelling, adaptive thresholding, quantified security, supervisory integration with a conventional relay, and validation realism. Because each cited study reports results on its own transformer model and dataset, the accuracy figures in Table 6.15 are the authors' own and are not directly comparable; the value of the table lies in the capability columns, which expose the structural gaps that the present work closes.

The capability matrix exposes a consistent pattern. Wavelet-plus-shallow-learning studies treat each analysis window independently and therefore discard the temporal evolution that distinguishes, for example, a decaying inrush DC component from a sustained internal fault; none integrates with a conventional relay, and security is at best reported implicitly. The convolutional approach of introduces limited local temporal awareness but still operates on fixed windows without long-range memory. The LSTM-based works correctly model temporal dependencies and report the highest standalone accuracies, yet they retain three structural weaknesses: the trip decision is issued by the network alone (so a single misclassification is a maloperation), the decision threshold is static, and security under out-of-distribution stress is not quantified. Adaptive-relay studies address thresholding but lack a modern sequence model and interpretability layer.

The proposed framework is, within the surveyed literature, the only scheme that simultaneously provides (i) explicit temporal modelling via a stacked attention-LSTM, (ii) a state-dependent adaptive threshold, (iii) a quantified security figure (100% on the test set, with graceful degradation characterised across five stress dimensions), and (iv) a supervisory handshake with a certified 87T element that guarantees the dependability floor. This combination is what converts a high-accuracy classifier into a deployable protection function: the LSTM contributes pattern-recognition power for security, while the classical relay contributes the deterministic, explainable dependability that protection standards require.

6.8.1 Discussion Relative to Specific LSTM Studies

The works most closely related to this thesis are the wavelet-LSTM schemes of Atiyah et al. and Alhamd et al. . Those studies established that an LSTM ingesting wavelet-derived features attains roughly 99.3–99.5% accuracy on simulated transformer data, validating the wavelet-plus-LSTM premise. The present work departs from them in three respects. First, the feature representation is reduced to a physics-informed six-dimensional energy vector—markedly more compact than the multi-component descriptors used previously—yet, as the ablation in Section 6.9 shows, this compact set sacrifices no discriminative power. Second, the LSTM is not the final arbiter: its output is gated by the adaptive 87T element, eliminating the false trips that an isolated classifier inevitably produces under deep CT saturation and sympathetic inrush (Section 6.4.2). Third, the model is deployed back into the electromagnetic-transient simulator through ONNX and exercised in closed loop, rather than scored offline, providing a substantially more realistic end-to-end validation. The dual-DNN scheme of Key et al. achieves comparable accuracy but addresses only the binary inrush-versus-internal problem; the present four-class formulation additionally resolves external faults and normal operation, which is necessary for a complete protection function.

6.9 Ablation Studies

To isolate the contribution of each design choice, a series of ablations was conducted on the unified dataset, varying one factor at a time while holding all others at their selected values. Table 6.16 reports the results. Removing the attention layer reduces accuracy by 0.71 percentage points and, more importantly, degrades inrush–external discrimination, confirming that the network benefits from selectively weighting the post-inception time steps. Replacing the two-layer (128→64) stack with a single 64-unit layer costs 1.34 points, while the detail-only and approximation-only feature subsets confirm the dominance of the detail-energy features established by the mutual-information analysis. Crucially, no configuration that retains the detail-energy features falls below 100% internal-fault recall, indicating that dependability is robust to architectural simplification even where overall accuracy declines.

6.10 Mother-Wavelet and Decomposition-Level Sensitivity

The selection of the db4 wavelet and three-level decomposition was validated by a systematic grid search over candidate wavelet families and depths, evaluated by overall accuracy and by the Fisher discriminant ratio of the resulting energy features. Table 6.17 summarises the comparison. The db4 wavelet attains the best accuracy–compactness trade-off: its four vanishing moments and jagged support match fault-inception transients, whereas smoother wavelets (sym, coif) disperse transient energy and the shorter Haar (db1) lacks the frequency selectivity to separate the harmonic bands. Decomposition beyond three levels yields single coefficients at the deepest scale for the 16-sample window, providing no additional energy information, consistent with the dyadic analysis of Section 5.4.2.

These results corroborate the design rationale of Chapter 5 and align with the prior wavelet-selection conclusions of and , which likewise identified the Daubechies family as well suited to transformer transient analysis, while extending those findings to the compact energy-feature, temporal-model setting adopted here.

6.11 ROC and Confidence-Threshold Analysis

Because the hybrid decision depends on the confidence threshold θ_conf, its selection was guided by a receiver-operating-characteristic (ROC) analysis on the validation set. Treating internal-fault detection as the positive class, the true-positive and false-positive rates were swept across confidence thresholds from 0.50 to 0.99. The resulting area under the ROC curve (AUC) for the standalone classifier is 0.9991, indicating near-perfect separability. Table 6.18 reports the dependability/security trade-off at representative thresholds, and Figure 6.10 plots the ROC curve. The selected operating point θ_conf = 0.85 maximises security without sacrificing the 100% dependability provided structurally by the 87T element, and lies on the flat upper-left region of the curve where performance is insensitive to small threshold perturbations.

6.12 Reliability and Economic Impact

The protection performance reported above translates into quantifiable reliability and economic benefits. Eliminating false trips on the test population removes the unnecessary service interruptions, customer-minutes lost, and transformer de-energisation cycles that accompany spurious operations; in multi-transformer substations, sympathetic-inrush false trips are a recognised cause of cascading disconnection, so the 100% security of the hybrid scheme directly improves substation availability. On the dependability side, the sub-13.1 ms clearing time reduces the through-fault energy $\int i^2(t)\,dt$ absorbed by the winding during an internal fault relative to conventional 30–40 ms harmonic-restraint relays, lowering insulation ageing and the mechanical forces that drive winding deformation. For a 300 MVA transmission transformer, avoiding a single major fault escalation or an unnecessary outage carries an economic value commonly estimated in the millions of dollars, so even a modest reduction in maloperation frequency yields a favourable cost–benefit balance for the proposed approach.

Equally important for adoption is the interpretability of the scheme. Each of the six input features maps to a specific frequency band and phase, the attention weights expose which time steps drive a decision, and the final trip requires the concurrence of a certified classical element. This transparency addresses a principal barrier to the deployment of machine learning in safety-critical protection: the ability of a protection engineer to understand, audit, and trust the relay’s behaviour.

6.13 Mapping of Results to Research Objectives

Across all four objectives the evidence is consistent: the framework attains the targeted dependability and security simultaneously, the feature and architecture choices are justified by controlled ablations rather than asserted, and the robustness and comparative analyses position the contribution clearly against both re-implemented baselines and the published literature.

6.14 Threats to Validity

In the interest of scientific rigour, the principal threats to the validity of the reported results are stated explicitly, together with the mitigations applied.

6.14.1 Internal Validity

The chief internal threat is information leakage between training and test partitions. This is mitigated by stratified splitting performed at the level of independent simulation cases—never at the level of individual sliding windows—so that windows from a single event never appear in more than one partition. Feature normalisation statistics (μ, σ) are computed exclusively on the training subset and then applied unchanged to validation and test data, preventing test-set leakage through the preprocessing pipeline. The 5-fold cross-validation of Section 6.3.4, with zero variance in internal-fault recall across folds, provides further assurance that the headline figures are not an artefact of a fortunate split.

6.14.2 External Validity

The dominant external threat is the simulation-to-field gap. All data originate from a single 300 MVA, 230/11 kV, Yd1 model, so generalisation to other ratings, vector groups, and core materials is unverified, and unmodelled phenomena—manufacturing tolerances, ambient and ageing effects, and measurement-chain idiosyncrasies—may shift the real-world feature distribution. Three measures partially mitigate this threat: physically grounded component models (nonlinear core and CT saturation with remanence); a merging-unit conditioning chain that injects realistic noise and quantisation; and an out-of-distribution stress programme spanning point-on-wave angle, remanent flux, frequency, noise, and fault resistance. Nonetheless, validation on field-recorded COMTRADE data and hardware-in-the-loop testing, as recommended in Chapter 7, remains essential before operational deployment.

6.14.3 Construct and Statistical Validity

The metrics adopted—dependability, security, accuracy, F1, AUC, and clearing time—are the standard constructs for protection evaluation and map directly to the engineering requirements stated in Chapter 1, supporting construct validity. The balanced test set of 1,920 cases (480 per class) is large enough that the reported per-class rates carry narrow confidence intervals, and the consistency between the independent test evaluation (Section 6.3) and the cross-validation (Section 6.3.4) argues against statistical over-fitting. Where reported accuracies from the external literature are cited (Tables 2.1 and 6.15), they are explicitly flagged as non-commensurable because each derives from a different dataset and protocol; only the re-implemented benchmark of Table 6.14 supports a like-for-like statistical comparison.

CHAPTER 7 CONCLUSION AND FUTURE SCOPES

7.1 Conclusion

This thesis has presented the design, development, and comprehensive evaluation of a hybrid adaptive transformer differential protection (HATDP) framework that synergistically combines a Discrete Wavelet Transform–Long Short-Term Memory (DWT-LSTM) classifier with a conventional 87T differential relay. The research addresses the persistent challenge in power transformer protection—the inability of conventional harmonic restraint (CHR) methods to simultaneously achieve high dependability and high security under adverse conditions including current transformer (CT) saturation, high-impedance faults, and complex inrush phenomena.

The MATLAB/Simulink environment was used to develop a detailed power-system model of a 300 MVA, 230 kV/11 kV power transformer with Yd1 vector group, including realistic representations of magnetization characteristics, CT saturation and remanence, and transmission-line parameters. Three-phase differential currents were sampled at 1.6 kHz (32 samples per cycle), and a comprehensive dataset of 12,800 event cases was generated across four operating conditions: normal load, external faults, magnetizing inrush, and internal faults. The signal-processing pipeline employs a db4 DWT on half-cycle sliding windows (16 samples), computing six energy features per classification instant—approximation energy and detail energy for each of the three phases. The two-layer LSTM classifier (128→64 hidden units) with global temporal attention, trained offline in Python using PyTorch, achieves 99.27% overall accuracy with 100% internal-fault recall and was exported to ONNX format (Opset 14) for integration into Simulink via the Deep Learning Toolbox Predict block, with inference latency of less than 0.74 ms per classification. The hybrid veto/AND-gate architecture achieves simultaneous 100% dependability (zero false negatives) and 100% security (zero false trips) on the 1,920-case test dataset, with a sub-cycle fault-clearing time of less than 13.1 ms—resolving the fundamental dependability–security trade-off that has constrained transformer differential protection for decades.

7.2 Key Findings

The principal findings of this research are summarized in three areas: feature robustness, hybrid security enhancement, and real-time feasibility.

7.2.1 Feature Robustness

Six DWT energy features—approximation energy and detail energy from the db4 decomposition of each phase's differential current—provide sufficient discriminative information for reliable four-class classification. The detail-energy features (100–800 Hz band) achieve an inter-class separation ratio exceeding 500 between internal faults and normal operation, while the approximation-energy features (0–100 Hz band) provide critical secondary discrimination between inrush and external-fault conditions. The LSTM classifier achieves 99.27% accuracy—substantially exceeding Wavelet-ANN (98.65%) and Wavelet-SVM (98.23%)—and maintains 100% internal-fault recall across all adversarial conditions, including CT saturation at all severity levels, high-impedance faults up to 300 Ω, noise down to 20 dB SNR, frequency deviations across the

47–53 Hz range, and extreme CT remanent flux up to ±95%. This consistent zero-false-negative rate provides the dependability foundation upon which the hybrid architecture is built.

7.2.2 Enhanced Security of Hybrid Architecture

The standalone 87T relay produces 77 false trips on the 1,920-case test dataset—primarily during external faults with CT saturation (51 false trips) and sympathetic-inrush events (26 false trips). The LSTM's supervisory veto blocks all 77 false trips (100% effectiveness), and in the hybrid AND-gate configuration—where both the 87T relay and the LSTM must agree to trip—the false-trip count drops to zero on the test dataset. For sympathetic inrush specifically, where the standalone 87T relay with cross-blocking produces 26 false trips (10.0% false-trip rate), the hybrid AND gate eliminates all false trips, achieving 100% security. This has significant practical implications, as sympathetic inrush is a common cause of false differential trips in multi-transformer substations.

7.2.3 Real-Time Feasibility

The complete hybrid protection pipeline executes in real time within the MATLAB/Simulink co-simulation environment. The end-to-end clearing-time budget comprises: data buffering 5.0 ms (quarter-cycle, 8 samples at 1.6 kHz); DWT decomposition and energy computation < 1.0 ms; LSTM inference (ONNX Predict block) < 0.74 ms; adaptive 87T relay computation < 1.0 ms; hybrid veto/AND-gate logic < 0.1 ms; for a total of less than 13.1 ms from fault inception to trip output. This is firmly sub-cycle at 50 Hz (one cycle = 20 ms) and improves on conventional CHR relays (30–40 ms) and modern digital relays (15–30 ms). The stateless LSTM inference design reduced latency by approximately 35% compared to a stateful implementation, while the MATLAB → Python (training) → ONNX → Simulink (validation) pipeline provides a validated pathway for transitioning AI-based protection from research to deployable relay firmware.

7.3 Research Outcomes

Immediate practical impact. The hybrid architecture demonstrates that 100% dependability and 100% security can be achieved simultaneously, challenging the long-held assumption that these objectives must be traded off. The validated MATLAB → Python (PyTorch training) → ONNX → Simulink (validation) pipeline provides a reproducible methodology that can be adopted by relay manufacturers. Sub-cycle clearing (< 13.1 ms) significantly reduces I^2t thermal stress on transformer windings. The six-energy-feature representation ensures physical interpretability—each feature maps to a specific frequency band and phase—addressing a key barrier to ML adoption in safety-critical applications.

Longer-term significance. The veto/AND-gate concept serves as a template applicable to transmission-line, busbar, and generator protection. The framework demonstrates a principled approach to integrating machine learning into protection systems that preserves the reliability and interpretability requirements of the protection community, contributing to the broader evolution toward digital substations and intelligent electronic devices.

7.4 Limitations of the Study

The LSTM model was trained exclusively on data from a 300 MVA, 230/11 kV, Yd1 transformer; generalization to other ratings, vector groups, or core materials has not been validated.

The dataset does not include inter-turn faults, which produce low-magnitude differential currents comparable to normal load and remain an open detection challenge.

Trip-only detection rate decreases for high-impedance faults above 300 Ω, and performance beyond 500 Ω has not been tested.

All results are simulation-based; validation with field-recorded COMTRADE data or hardware-in-the-loop (HIL) testing is essential before practical deployment.

The simulation model represents a fixed power-system topology; performance under significant post-contingency topological changes has not been evaluated.

7.5 Recommendations for Future Work

7.5.1 Hardware-in-the-Loop Testing

The hybrid relay algorithm—including DWT computation, ONNX-based LSTM inference, 87T relay logic, and hybrid decision—should be implemented on a real-time hardware platform (e.g., Speedgoat, OPAL-RT, or FPGA/ARM-based numerical relay) and tested against a real-time digital simulator. The HIL testbed should evaluate deterministic execution time, ADC quantization effects, CT burden and secondary-wiring impedance impacts, and long-duration stability.

7.5.2 Inter-Turn Fault Detection

A detailed multi-winding transformer model should be developed to simulate inter-turn faults with varying turn ratios (1%–10% of winding turns), capturing the asymmetrical magnetic-flux distribution and high-frequency transients characteristic of such faults. The DWT feature pipeline should be evaluated for inter-turn fault sensitivity, with additional features investigated if the existing six-feature representation proves insufficient.

7.5.3 Field Deployment and IEC 61850 Integration

Collaboration with electric utilities should be established to obtain COMTRADE fault records from operational 300 MVA-class transformers for benchmarking. Trip commands and supervisory alarms should be mapped to IEC 61850 GOOSE messages, and the integration should address IEC 61850-9-2 Sampled-Values requirements including network latency, packet loss, and data-synchronization challenges.

7.5.4 Transfer Learning for Multi-Configuration Deployment

Transfer learning should be explored to enable knowledge transfer from the 300 MVA model to other transformer configurations with minimal additional training data (500–1,000 configuration-specific events). Fine-tuning of the pre-trained LSTM layers could significantly reduce data-collection and training effort. An online-learning mechanism with model-drift detection and safeguards against catastrophic forgetting should also be investigated.

7.6 Concluding Remarks

This thesis has demonstrated that a hybrid architecture combining a conventional adaptive 87T relay with a DWT-LSTM classifier through a supervisory veto/AND gate achieves what neither element can achieve alone: simultaneous 100% dependability and 100% security with sub-cycle fault-clearing time validated in the MATLAB/Simulink real-time co-simulation environment. The key design philosophy treats the LSTM not as a replacement for the conventional relay but as an intelligent supervisory layer that enhances security by vetoing false trips, while the relay provides the safety-certified first line of defence that the protection-engineering community trusts.

The complete MATLAB/Simulink → Python/PyTorch → ONNX → Simulink pipeline demonstrates a practical, reproducible workflow bridging AI research and protection-engineering practice. While the simulation-based nature of the validation represents a limitation, the zero-false-negative guarantee combined with the dramatic reduction in false trips and sub-cycle clearing time provides compelling evidence that the hybrid DWT-LSTM + 87T architecture is a viable direction for next-generation transformer differential protection.

7.7 Broader Outlook

The hybrid philosophy demonstrated in this thesis—pairing a learned, security-enhancing supervisor with a certified, dependability-guaranteeing classical element—generalises beyond transformer differential protection. The same handshake structure can be applied to busbar protection, where CT saturation during external faults similarly threatens security; to transmission-line distance and differential protection, where evolving and high-resistance faults challenge conventional settings; and to generator protection, where the discrimination between magnetising phenomena and genuine faults is analogously difficult. In each case, the learned component supplies pattern-recognition power while the classical element supplies the deterministic behaviour that protection standards demand, and the explicit gate keeps the failure modes of the two components from compounding.

As power systems evolve toward inverter-dominated generation, the waveform signatures on which conventional relays depend will continue to drift away from the sinusoidal, harmonic-rich assumptions of legacy algorithms. Data-driven supervisors that learn directly from the prevailing signal statistics, retrained or fine-tuned as the system changes, offer a principled path to maintaining protection security under these shifting conditions. The transfer-learning and online-adaptation directions outlined in Section 7.5.4 are therefore not merely incremental refinements but enabling capabilities for protection in a low-inertia, converter-rich grid. Coupled with the digital-substation integration of Section 7.5.3, they point toward a generation of intelligent electronic devices in which certified classical logic and continuously adapting learned supervision operate together as a matter of course.

In summary, this thesis contributes not only a specific, validated transformer-protection scheme but also a transferable design pattern for the responsible integration of machine learning into safety-critical protection. By keeping the learned model in a supervisory role, quantifying both dependability and security, stress-testing well beyond the training distribution, and validating in closed-loop co-simulation, the work offers a template that the protection community can adopt with confidence as it navigates the transition to intelligent, digital, and increasingly autonomous power systems.

LIST OF PUBLICATIONS

The following publications have been derived from the research work presented in this thesis:

Author Name, Supervisor Name, and Co-Author Name, “Hybrid Adaptive Transformer Differential Protection Using a DWT-LSTM Intelligent Framework with Enhanced CT Saturation and Noise Robustness,” IEEE Access, vol. XX, pp. XXXXX–XXXXX, 2025. DOI: 10.1109/ACCESS.2025.XXXXXXX.

Summary: This journal paper presents the complete DWT-LSTM framework for power transformer differential protection, including the three-level db4 wavelet-decomposition methodology, the compact six-dimensional energy feature-extraction pipeline, and the LSTM classification architecture with global temporal attention. The paper reports an overall classification accuracy of 99.27%, with 100% dependability and 100% security in the hybrid configuration and a sub-cycle operating time below 13.1 ms for internal faults. Comprehensive evaluation under CT saturation, measurement noise, high-impedance faults, and varying magnetizing conditions is provided, along with comparative analysis against seven existing methods.

Author Name, Supervisor Name, and Co-Author Name, “Multi-Resolution Feature Extraction for Transformer Protection: A Discrete Wavelet Transform Approach with Mutual Information-Based Feature Selection,” Proceedings of the IEEE International Conference on Power Systems (ICPS), pp. XXX–XXX, 2025. DOI: 10.1109/ICPS.XXXX.XXXXXXX.

Summary: This conference paper focuses on the DWT-based feature-extraction methodology. It presents a systematic analysis of mother-wavelet selection (comparing db4, db6, sym4, sym5, coif3, and bior2.2), decomposition-level optimization, and energy feature extraction from wavelet coefficients. A mutual-information-based feature-importance ranking is introduced, demonstrating that the six energy features—with the three detail-energy components capturing 72.3% of the discriminative information—are sufficient for reliable four-class discrimination. Ablation-study results on feature-subset size are also presented.

Author Name, Supervisor Name, and Co-Author Name, “Comparative Analysis of Intelligent Methods for Transformer Differential Protection Under Challenging Operating Conditions,” International Journal of Electrical Power & Energy Systems, vol. XXX, pp. XXXXX–XXXXX, 2025. DOI: 10.1016/j.ijepes.2025.XXXXXX.

Summary: This journal paper provides a comprehensive comparative analysis of seven transformer differential protection methods, including conventional harmonic restraint, wavelet-ANN, wavelet-SVM, wavelet-PNN, wavelet-CNN, and deep-learning approaches. The comparison is conducted under a unified simulation framework covering five challenging operating conditions: CT saturation, measurement noise, high-impedance faults, severe inrush conditions, and combined worst-case scenarios. The proposed DWT-LSTM framework is shown to outperform all compared methods across the evaluated metrics.